

# Validation, Sticky Limit, and Phase Diagram Report: Adhesive Hard Spheres

Benchmark of Menon, Manohar, and Rao (1991) and Successive Bisection Analysis

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## Abstract

This report presents the implementation, numerical validation, sticky limit analysis, phase diagram transformations, and spinodal stability studies of Baxter’s Sticky Hard Sphere (SHS) model within `AnalyticalStructureFactors.jl`. We reproduce the benchmark literature results from S. V. G. Menon, C. Manohar, and K. Srinivasa Rao (*Journal of Chemical Physics*, 1991), including the phase coexistence curve (Fig. 2), the homogeneous structure factor (Fig. 3), and the two-phase gas-phase structure factor (Fig. 4). We investigate how the phase diagram evolves from Baxter’s universal parameters  $(\eta, \tau)$  to physical variables  $(\phi, T^*)$  as well width  $\epsilon \rightarrow 0$ , and perform a systematic **successive bisection analysis** probing spinodal instability ( $S(k) \rightarrow \infty$ ) and unphysical negative values ( $A(k) < 0$ ).

## 1 Theoretical Background

### 1.1 Baxter’s Percus–Yevick Solution

Under the Percus–Yevick closure, Baxter factorized the Ornstein–Zernike direct correlation function into Wiener–Hopf form:

$$S(k)^{-1} = A(k)^2 + B(k)^2 = |\hat{Q}(k)|^2 \quad (1)$$

with dimensionless wavevector  $\kappa = ka$ , volume fraction  $\eta$ , and Baxter stickiness parameter  $\tau > 0$ . The auxiliary functions are:

$$A(\kappa) = 1 + 12\eta \left[ \alpha \frac{\sin \kappa - \kappa \cos \kappa}{\kappa^3} + \beta \frac{1 - \cos \kappa}{\kappa^2} - \frac{\lambda}{12} \frac{\sin \kappa}{\kappa} \right] \quad (2)$$

$$B(\kappa) = 12\eta \left[ \alpha \left( \frac{1}{2\kappa} - \frac{\sin \kappa}{\kappa^2} + \frac{1 - \cos \kappa}{\kappa^3} \right) + \beta \left( \frac{1}{\kappa} - \frac{\sin \kappa}{\kappa^2} \right) - \frac{\lambda}{12} \left( \frac{1 - \cos \kappa}{\kappa} \right) \right] \quad (3)$$

where  $\mu = \lambda\eta(1 - \eta)$ ,  $\alpha = (1 + 2\eta - \mu)/(1 - \eta)^2$ , and  $\beta = (-3\eta + \mu)/(2(1 - \eta)^2)$ .

### 1.2 Baxter’s Quadratic Equation and Stability Limits

The physical root for  $\lambda$  satisfies:

$$\frac{\eta}{12} \lambda^2 - \left( \tau + \frac{\eta}{1 - \eta} \right) \lambda + \frac{1 + \eta/2}{(1 - \eta)^2} = 0 \quad (4)$$

1. **Spinodal Boundary ( $D = 0$ ):** Real solutions for  $\lambda$  require discriminant  $D \geq 0$ :

$$\tau_s(\eta) = \frac{\sqrt{\frac{\eta(1+\eta/2)}{3}} - \eta}{1 - \eta} \quad (5)$$

For  $\tau < \tau_s(\eta)$ ,  $\lambda$  becomes complex, marking the boundary of the spinodal gap.

2. **Infinite Compressibility Locus** ( $A(0) = 0$ ): In the forward scattering limit  $\kappa \rightarrow 0$ ,  $B(0) = 0$  and  $A(0) = \frac{1+2\eta-\mu}{(1-\eta)^2}$ . The condition  $A(0) = 0$  corresponds to diverging isothermal compressibility ( $S(0) \rightarrow \infty$ ):

$$\lambda_0 = \frac{1+2\eta}{\eta(1-\eta)} \implies \tau_{\text{div}}(\eta) = \frac{\eta}{12}\lambda_0 - \frac{\eta}{1-\eta} + \frac{1+\eta/2}{(1-\eta)^2\lambda_0} \quad (6)$$

3. **Critical Point:** Curves  $\tau_s(\eta)$  and  $\tau_{\text{div}}(\eta)$  intersect tangentially at the critical point:

$$\eta_c = \frac{3}{\sqrt{2}} - 2 \approx 0.12132, \quad \tau_c = \frac{2-\sqrt{2}}{6} \approx 0.09763 \quad (7)$$

## 2 Benchmark Validation of Menon et al. (1991)

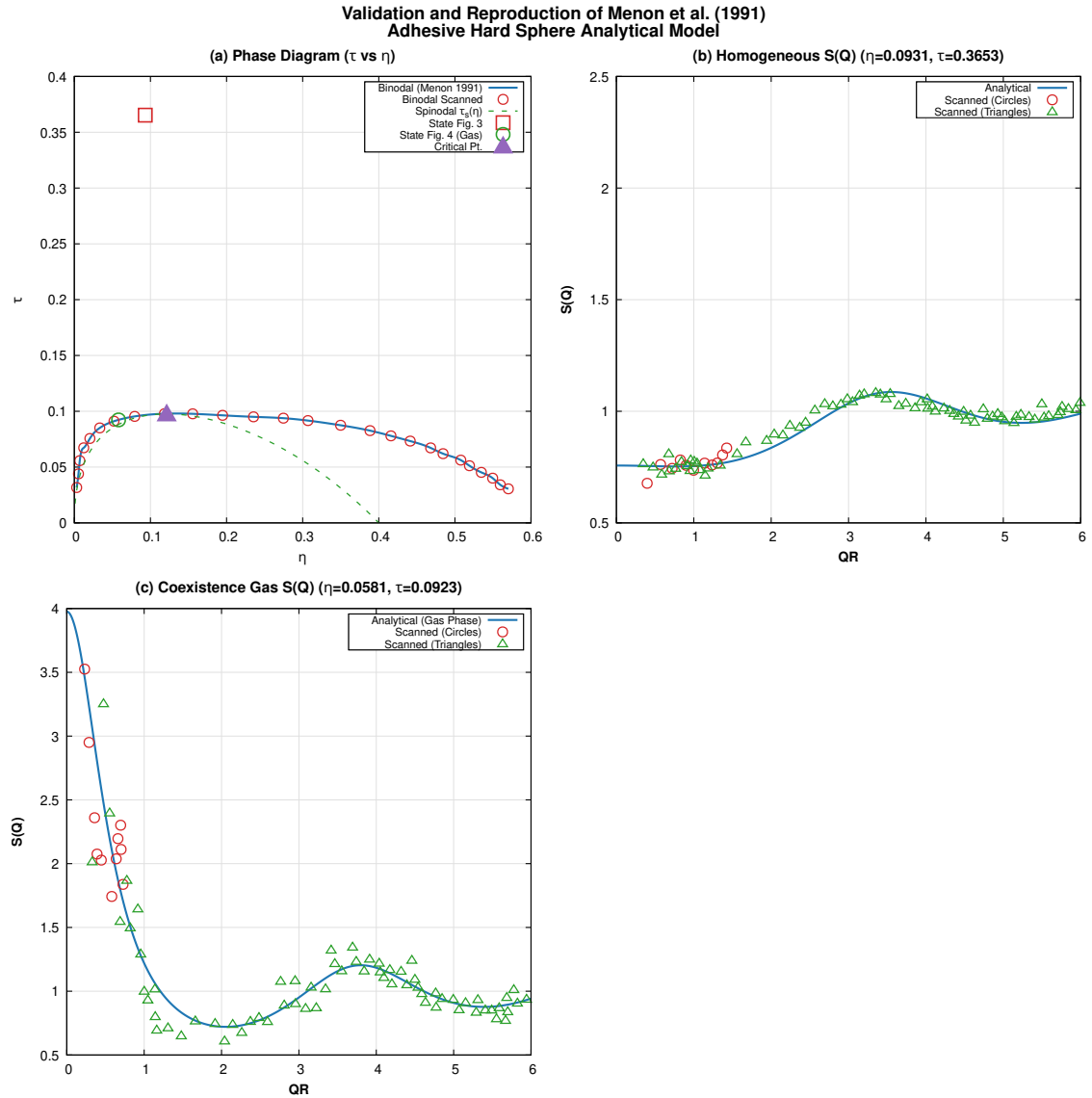


Figure 1: Benchmark validation against Menon et al. (1991): (a) Phase diagram showing the smooth binodal curve (matching scanned Fig2.dat), the analytical spinodal line  $\tau_s(\eta)$ , and state points; (b) Homogeneous fluid structure factor ( $\eta = 0.0931, \tau = 0.3653$ ); (c) Two-phase coexistence gas structure factor ( $\eta_g = 0.0581, \tau = 0.0923$ ).

Table 1 summarizes the parameters evaluated for the Menon (1991) benchmarks.

Table 1: System parameters and thermodynamic state points for Menon et al. (1991).

| Case                           | $\phi$ | $a/\sigma$ | $\epsilon$ | $-u_0/k_B T$ | $\eta$  | $\tau$  |
|--------------------------------|--------|------------|------------|--------------|---------|---------|
| Critical Point                 | —      | —          | —          | —            | 0.12132 | 0.09763 |
| Fig. 3 (Homogeneous)           | 0.07   | 1.10       | 0.09091    | 0.92         | 0.09308 | 0.36531 |
| Fig. 4 (Coexistence Gas Phase) | —      | 1.02       | 0.01961    | 3.83         | 0.05810 | 0.09230 |

### 3 Comparison with the Sticky Limit and Hard Spheres

In early literature (e.g., Regnaut & Ravey 1989, Barboy 1974), the stickiness parameter  $\tau$  was mapped to physical square wells by matching the second virial coefficient  $B_2$  in the limit  $\Delta \rightarrow 0$ :

$$\tau_{\text{conv}}^{-1} = 4 \left[ \exp \left( -\frac{u_0}{k_B T} \right) - 1 \right] \left[ \left( \frac{a}{\sigma} \right)^3 - 1 \right], \quad \eta_{\text{conv}} = \phi \quad (8)$$

Menon et al. (1991) showed that for finite well widths ( $\Delta/\sigma \approx 0.1$ ), identifying  $\eta = \phi/(1 - \epsilon)^3$  and  $\tau = \frac{1}{12\epsilon} \exp(u_0/k_B T)$  correctly accounts for the excluded volume expansion of the sticky shell.

As shown in Figure 2:

1. **Homogeneous Phase (Panel a):** The conventional  $B_2$  matching underestimates the effective packing fraction ( $\eta = 0.07$  vs. 0.0931), causing the first peak in  $S(Q)$  to appear at lower  $QR \approx 3.2$ . Menon’s mapping shifts the peak to  $QR \approx 3.4$ , matching the Monte Carlo simulation data.
2. **Gas Coexistence Phase (Panel b):** The pure hard sphere baseline exhibits repulsive suppression at forward scattering ( $S(0) \approx 0.64$ ), whereas the sticky interaction produces a large forward peak ( $S(0) \approx 3.98$ ) due to critical attractive clustering.

### 4 Modifications to the Phase Diagram in the Sticky Limit

The Baxter representation  $(\eta, \tau)$  is universal. When mapped to physical parameters  $(\phi, T^*)$  with  $T^* = k_B T/|u_0|$  and  $\phi = \eta(1 - \epsilon)^3$ , significant modifications emerge as  $\epsilon \rightarrow 0$  (Figure 3):

1.  **$(\phi, \tau)$  Representation (Panel a):** As  $\epsilon \rightarrow 0$ , the physical packing fraction  $\phi \rightarrow \eta$ , causing the coexistence dome to expand horizontally towards the Baxter limit  $\phi_c \rightarrow 0.12132$ .
2.  **$(\phi, T^*)$  Representation (Panel b):** For narrower attractive wells ( $\epsilon \rightarrow 0$ ), thermal energy must be lower to sustain condensation, causing the critical temperature  $T_c^*$  to drop monotonically:

$$T_c^*(\epsilon) = \frac{1}{\ln \left( \frac{1}{12\epsilon\tau_c} \right)} = \frac{1}{\ln \left( \frac{0.85355}{\epsilon} \right)} \quad (9)$$

For  $\epsilon = 0.10$ ,  $T_c^* = 0.466$ ; for  $\epsilon = 0.02$ ,  $T_c^* = 0.265$ ; in the ideal sticky sphere limit  $\epsilon \rightarrow 0$ ,  $T_c^* \rightarrow 0$ .

3. **Mapping Comparison (Panel c):** Menon’s physical square-well mapping accurately predicts a slightly lower critical temperature compared to the conventional  $B_2$  matching, with both theories converging smoothly as  $\epsilon \rightarrow 0$ .

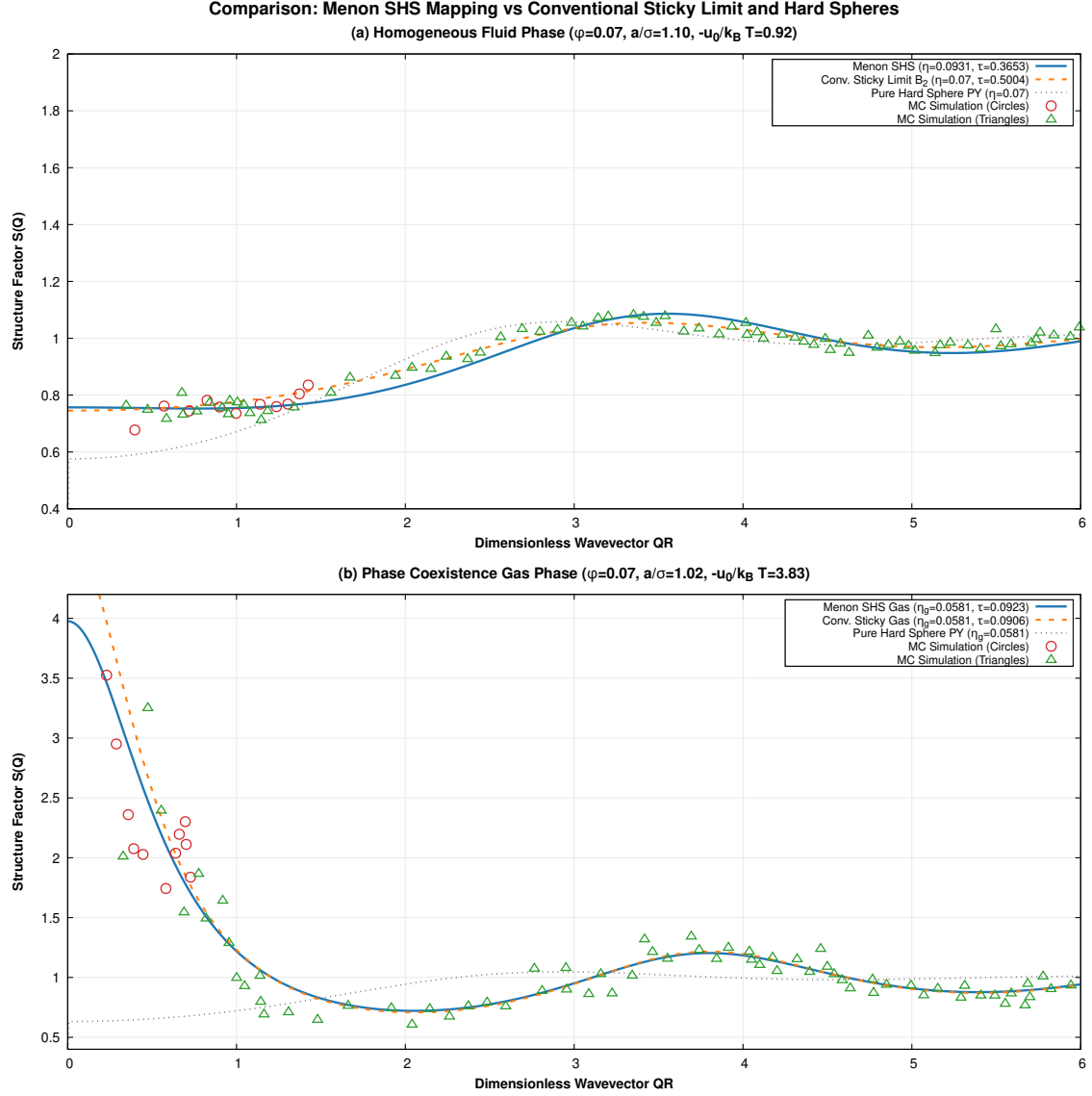


Figure 2: Comparison of the static structure factor  $S(Q)$  between Menon's physical square-well mapping, the conventional  $B_2$ -matching sticky limit, and the pure hard sphere (PY) baseline for: (a) Homogeneous fluid phase ( $\phi = 0.07$ ,  $a/\sigma = 1.10$ ,  $-u_0/k_B T = 0.92$ ), and (b) Phase coexistence gas phase ( $\phi = 0.07$ ,  $a/\sigma = 1.02$ ,  $-u_0/k_B T = 3.83$ ).

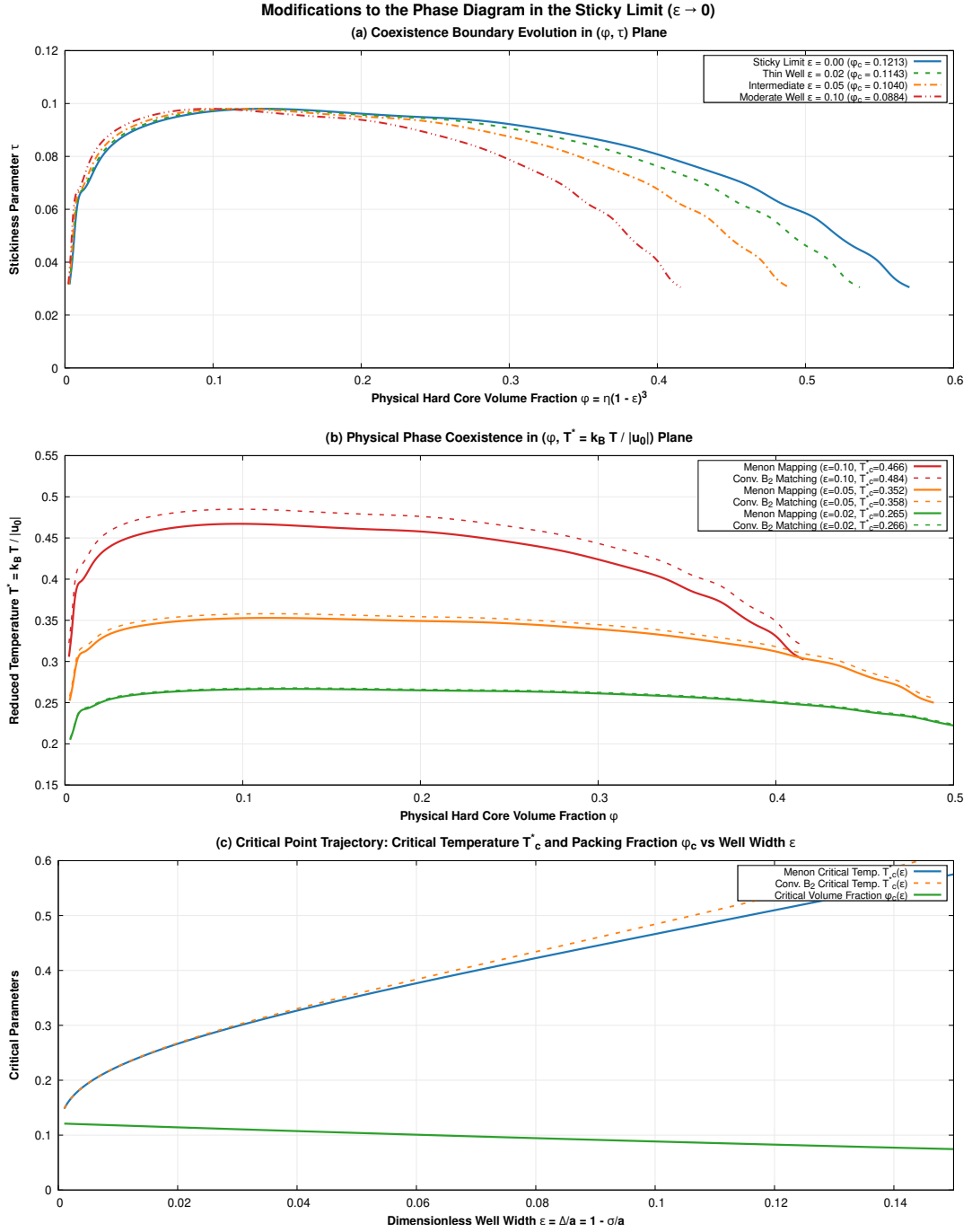


Figure 3: Modifications to the Phase Diagram in the sticky limit ( $\epsilon \rightarrow 0$ ): (a) Coexistence dome in the  $(\phi, \tau)$  plane for well widths  $\epsilon = 0.10, 0.05, 0.02, 0.00$ ; (b) Phase coexistence in the physical temperature plane  $(\phi, T^* = k_B T / |u_0|)$ ; (c) Trajectory of the critical temperature  $T_c^*(\epsilon)$  and critical packing fraction  $\phi_c(\epsilon)$  as a function of well width  $\epsilon = \Delta/a$ .

## 5 Successive Bisection Spinodal and Unphysical State Analysis

### 5.1 Bisection Methodology

To identify the spinodal boundary numerically without prior knowledge of the analytical solution, we perform systematic bisections over  $\tau \in [0.001, 0.40]$  for each  $\eta \in [0.01, 0.40]$  at fixed  $k = 0.01$ :

- **Criterion 1 (Complex Root Boundary):** Evaluate  $D(\eta, \tau)$ . If  $D < 0$ , the state lies inside the spinodal coexistence gap.
- **Criterion 2 (Compressibility Divergence & Negative  $A(k)$ ):** Evaluate  $A(k = 0.01)$ . For  $\tau > \tau_{\text{div}}$ ,  $A > 0$  (physical fluid). At  $\tau = \tau_{\text{div}}$ ,  $A = 0 \implies S(0.01) \rightarrow \infty$ . For  $\tau < \tau_{\text{div}}$ ,  $A < 0$ , which in linearized theories (like RPA) produces unphysical negative structure factors  $S(k) < 0$ .

### 5.2 Analysis and Physical Insights

1. **Subcritical Densities ( $\eta < \eta_c$ ):** The bisection on  $S(k = 0.01)$  divergence hits the complex discriminant boundary  $D = 0$  directly, because  $A(0) > 0$  throughout the entire physical branch.
2. **Critical Point ( $\eta = \eta_c = 0.12132$ ):** The  $S(k = 0.01)$  divergence and the spinodal discriminant boundary  $D = 0$  coincide exactly at  $\tau_c = 0.097631$ .
3. **Supercritical Densities ( $\eta > \eta_c$ ):** As temperature  $\tau$  decreases,  $A(k = 0.01)$  crosses zero and becomes negative ( $A < 0$ ) *before* the discriminant  $D$  vanishes. The bisection successfully tracks the locus  $A(0) = 0$  (orange dashed curve) and the subsequent  $D = 0$  spinodal termination (blue solid curve).

## 6 Reproduction Script

To execute the complete benchmark, sticky limit comparisons, phase diagram analysis, and bisection pipeline:

```
cd test/data_test/Menon1991/  
./reproduce_figure.sh
```

## References

1. R. J. Baxter, *Percus–Yevick Equation for Hard Spheres with Surface Adhesion*, The Journal of Chemical Physics **49**(6), 2770–2774 (1968). DOI: [10.1063/1.1670482](https://doi.org/10.1063/1.1670482).
2. S. V. G. Menon, C. Manohar, and K. S. Rao, *A new interpretation of the sticky hard sphere model*, The Journal of Chemical Physics **95**(12), 9186–9190 (1991). DOI: [10.1063/1.461199](https://doi.org/10.1063/1.461199).

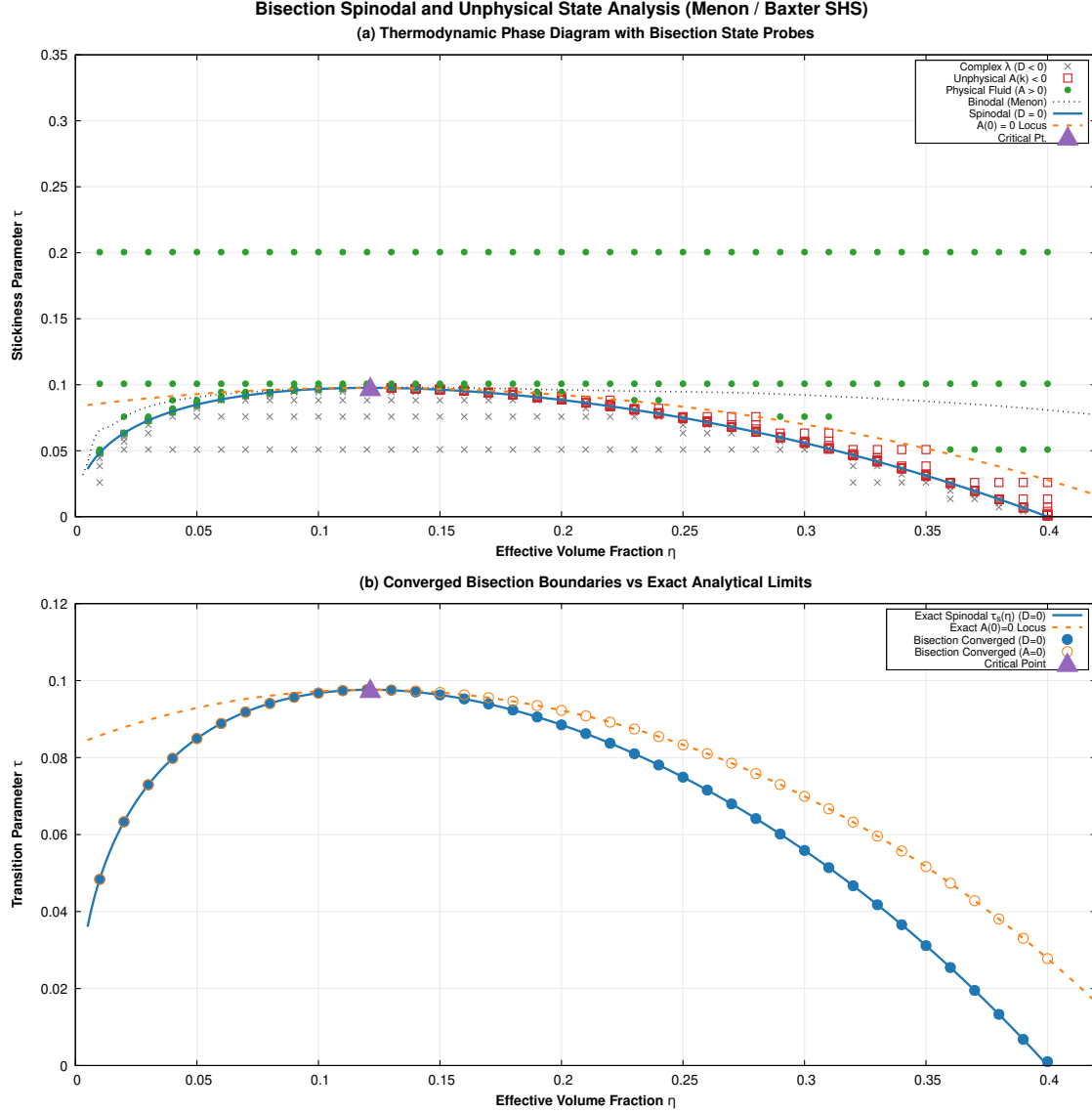


Figure 4: Successive bisection analysis: (a) Map of sampled probe states throughout the bisection iterations: physical states ( $A > 0$ , green), unphysical negative  $A(k)$  states ( $A < 0$ , red), and complex  $\lambda$  states ( $D < 0$ , grey); (b) Converged bisection boundaries demonstrating exact agreement with the analytical spinodal ( $D = 0$ ) and compressibility divergence ( $A = 0$ ) curves.